

Aliah University

End Semester Examination (Odd Semester) - 2022

Course: Civil Engineering,

Year: 3rdYear;

Subject: Soil Mechanics

Subject Code: CENUGPC14

Max. Marks: 80

Times: 3 hrs.

Answer all questions.

- Q.1 a) Write short notes on the following. (2X5=10)
(i) Major soil deposits of India.
(ii) Formation of clay minerals.
- b) Compare I.S. Light and Heavy compaction tests w.r.t. weight of rammer, height of free drop, number of layers and compactive energy imparted. (5)
- c) Derive the relationship between specific gravity of soil solids, void ratio, and bulk density of a saturated soil mass. (5)
- Q.2. (a) In a liquid limit test, the number of blows corresponding to 20% and 60% water contents were 100 and 10 respectively. Determine the flow index and liquid limit of the soil. (5)
- (b) A partially saturated sample from a borrow pit has a natural moisture content of 15 percent and bulk unit weight of 1.9 g/cc. The specific gravity of solid is 2.70. Determine the degree of saturation and void ratio. What will be the unit weight of the sample on saturation? (6)
- (c) The classification tests on a sample of soil from a highway project site reveal that 60 percent of the soil passes US sieve no 200. The liquid limit of the soil is 38 percent and plastic limit is 25 percent. Calculate the group index of the soil. (4)
- (d) A concentrated load of 2000 kN is applied at the ground surface. Determine the vertical stress at point located at 6 m depth and at a horizontal distance of 5 m from the axis of the load. (5)
- Q.3. (a) What is flow net? What are the properties of the flow net? Derive an expression to estimate the seepage loss from a reservoir using flow net. (2+4+10)
- (b) A masonry dam has pervious sand as foundation. Determine the maximum permissible upward gradient, if a factor of safety of 4 is required against boiling? For the sand $n=45\%$ and $G=2.65$. (4)
- Q.4. (a) Define the terms: relative density, relative compaction and zero air voids density. How is compaction different from consolidation? (6+4)
- (b) In a falling head permeability test on sample 12.2 cm high and 44.41 cm² in cross-sectional area, the water level in stand pipe of 6.25 mm internal diameter dropped from a height of 75 cm to 24.7 cm in 15 minutes. Find the coefficient of permeability. (5)
- (c) Why seepage velocity is greater than discharge velocity? Discuss. (5)

ALIAH UNIVERSITY

End-Semester Examination- December, 2022

Year: 3rd Year (Dept. of Civil Engg.) Semester: 5th Semester
Subject: Open Channel Flow Subject Code: CENUGPC13
Full Marks: 80 Time: 3 hr.

Attempt all questions and assume suitable data if required.

1. (a) Discuss any two of the followings: 2x4 = 8
- I. Difference between Open channel flow and pipe flow.
 - II. Velocity and pressure distribution in open channel flow.
 - III. State of flow and regime of flow in open channel.
- (b) Derive expressions for the flow area, A , wetted perimeter, P , hydraulic radius, R , top-water surface width, B , and hydraulic depth, D , for the following channel cross sections: 2x4 = 8
- I. Trapezoidal (bottom width = B_0 , side slopes = 1 V : s H)
 - II. Triangular (side slopes = 1 V : s H)
2. (a) A 3.0 m wide rectangular channel carries a discharge of 1.85 m³/s at a depth of 0.5 m. A contraction of the channel width is required at a certain section. Find the greatest allowable contraction in the width for the upstream flow to be possible as specified. 8
- (b) Discuss the following: 2x4 = 8
- I. Transition in channel with hump.
 - II. Transition in channel with a change in width.
- (a) A trapezoidal channel of 3.0 m bed width and $m = 1.5$ carries a full supply of 10.0 m³/s at a depth of 1.5 m. What would be the discharge at half of full supply depth? What would be the depth at half of full supply discharge? 8
- (b) Show that the normal depth in a triangular channel of side slope m horizontal : 1 vertical, is given by 8
- $$y_0 = 1.1892 (Q \cdot n / \sqrt{s_0})^{3/8} ((m^2 + 1) \sqrt{m^5})^{1/8}$$
- where other notations carries their usual meaning.
4. Discuss the gradually varied flow profile classification and feature of flow profiles. 16
5. (a) Discuss the various types of jump and its characteristics. 8
- (b) In a hydraulic jump taking place in a rectangular channel the discharge intensity and head loss are found to be 4.7 m³/s/m and 6.0 m respectively. Determine the sequent depths of the jump. 8

Aliah University

End Semester Examination, December 2022

Course: B. Tech. (Civil Engineering)

Year & Semester: 3rd Yr. (V Sem.)

Subject: Design of Steel Structures

Code: CENUGPC12

Full Marks: 80

Time: 3hrs.

Answer any four questions. IS 800:2007 and Steel Table allowed. Assume suitable data if required. The yield and ultimate strengths of the steel are 250 MPa and 410 MPa, respectively

1. (i) Explain the following terms: Plastic moment, Plastic hinge, Plastic design and Shape factor.
(ii) Explain different failure modes of bolted joints. [10+10=20]
2. Design a single angle to carry a tension of 120kN. The end connection is to be done using M20 bolts of product Grade C and property Class 4.6. The yield and ultimate strengths of the steel are 250 MPa and 410 MPa, respectively. Assume that the bolt threads are outside the shear plane. [20]
3. Calculate the design load carrying capacity in compression of a discontinuous strut 2.5 m long (the centre to centre distance between the end connections) consisting of two equal angles 75×75×6 in the following cases:
 - (i) Connected to the same side of a gusset by more than one bolt in each angle.
 - (ii) Connected to both sides of 8 mm thick gusset by two bolts. The grade of steel is E250[12+8=20]
4. Design a built-up column of the effective length of 6.0 m to carry an axial load of 800 kN using single lacing. [20]
5. A simply supported beam of span 6 m carrying a distributed load of 20 kN/m (including self-weight) throughout the span. Design the beam assuming that it is laterally supported along its length. Also compute the design bending strength of the same beam, if the compression flange is laterally unsupported. Assume that full torsional and warping restraints are provided at the supports. Also, assume that the load acts on the upper flange which will have a destabilizing effect.

[12+8=20]

Aliah University
Department of Civil Engineering
Subject-Transportation Engineering
Subject code- [CENUGPC15]

Total marks: 80

Time: 3 hours

Group-A

1. Choose the correct alternatives for the following: $1 \times 10 = 10$

(i) The coefficient of friction between the tire and pavement considered for determining safe stopping distance, is in the range of

- (a) 0.07-0.15
- (b) 0.20-0.30
- ☒ (c) 0.35-0.40
- (d) 0.50-0.55

ii) The design speed recommended by IRC for National highways passing through rolling terrain is in the range of

- (a) 100-80 Kmph
- ☒ (b) 80-65 Kmph
- (c) 120-100 Kmph
- (d) 50-40 Kmph

iii) As per PIEV theory, Intellection time is

- (a) the time required for the sensations received by the eyes or ears
- ☒ (b) the time required for understanding the situation
- (c) the time elapsed during emotional sensations and disturbances
- (d) the time taken for final action

iv) Minimum length of overtaking zone

- a) $2 \times (\text{Overtaking Sight Distance})$
- b) $5 \times (\text{Overtaking Sight Distance})$
- ☒ c) $3 \times (\text{Intermediate sight distance})$
- d) None of these.

v) The psychological widening is given by the formula:

- a) $V^3/9.5\sqrt{R}$
- ☒ b) $V/9.5\sqrt{R}$
- c) $V/9.5R$
- d) $V^3/9.5\sqrt{R}$

vi) In pay as you go method, the payment for highway improvements, maintenance, and operation is made from

- a) Borrowed money
- ☒ b) Central revenue
- c) Future income
- d) None of these.

vii) The mechanical extra widening required for 10.5 m wide pavement on a horizontal curve of radius R meter is

- a) $L^2/2R$
- b) L^2/R
- c) $3L^2/2R$
- d) $2L^2/3R$

viii) The contact pressure is found to be more than tyre pressure when the tyre pressure

- a) is less than 7 kg/cm²
- b) is greater than 7 kg/cm²
- c) is less than 7 kg/cm²
- d) is greater than 9 kg/cm²

ix) ESWL at a depth of 2S is equal to

- (a) $P/2$
- (b) P
- (c) $2P$
- (d) $3P$

x) The critical combination of stress is for the corner region at mid night given by

- (a) $\sigma_{critical} = \text{Load stress} + \text{Warping stress}$
- (b) $\sigma_{critical} = \text{Load stress} + \text{Warping stress} + \text{frictional stress}$
- (c) $\sigma_{critical} = \text{Load stress} + \text{Warping stress} - \text{frictional stress}$
- (d) None of the above

Group-B

[Short Answer Type Questions]

Answer any eight of the following:

$$8 \times 5 = 40$$

2. Write down the components of the cross-section of a typical two-lane-divided highway.
3. Briefly discuss the engineering survey to finalize highway alignment and location.
4. A vertical summit curve is formed at the intersection of two gradients, +3.0 and -5.0 percent. Design the length of the summit curve to provide a stopping sight distance for a design speed of 80 Km/h. Assume other data.
5. Calculate the superelevation required on a road curve of 240 m radius with mixed traffic conditions. The design speed is 80 km/hr. The coefficient of friction is 0.15. The road passes through rolling terrain.
6. Explain "PIEV" theory.
7. On a two-way traffic road, the speed of overtaking and over taken vehicles is 80 and 50 Km/h, respectively. Suppose the acceleration of overtaking vehicle is 0.98 m/sec². Calculate safe overtaking sight distance.

8. Write short notes on various types of the gradient.
9. Write down the significance of the aggregate impact value test and the loss angles abrasion test.
10. Write down the functions of the dowel bar and tie bar.
11. Discuss expansion joints, contraction joints, and construction joints.

Group-C
[Long Answer Type Questions]

Answer any three of the following questions:

3x10

12. Calculate the length of the transition curve required for a carriageway of 7.5 m, width on straight portions, if the design speed is 65 km/hr and the road is passing through rolling terrain (Superelevation 1 in 150, pavement rotated about the center line). The radius of the horizontal curve is 200 m.
13. Derive an expression for calculating the overtaking sight distance on a highway.
14. What are the various tests carried out on bitumen? Briefly mention the principle and uses of each test.
15. A fixed time 2-phase signal is to be provided at an intersection having a North-South and an East-West road where only straight-ahead traffic is permitted. The design hour flows from the various arms and the saturation flows for these arms given in the following table:

	North	South	East	West
	h	h		t
Design hour flow (q) in PCU s/hour	800	400	750	1000
Saturation flow (s) in PCU s/hour	2400	2000	3000	3000

Calculate the optimum cycle time and green time for the minimum overall delay. The intergreen time should be the minimum necessary for efficient operation. The time lost per phase due to starting delays can be assumed to be 2 seconds. The value of amber period is 2 seconds. Sketch the timing diagram for each phase.

16. Write down the comparisons between flexible pavement and rigid pavement.

“UG” REGULAR EXAMINATION – Dec. 2022
3RD YEAR (5TH SEMESTER) CIVIL ENGINEERING
STRENGTH OF MATERIALS

MENUGOE03

Full Marks: 80

Time: 3 Hours

Question No. 1 is Compulsory. Answer any five from the rest.

1. Choose the correct answer from the given alternatives.

[5 X 2=10]

- (i) On two mutually perpendicular planes, normal stresses are **120 MPa** and **60 MPa**. Also shear stress are **40 MPa**. What is the value of maximum shear stress at that point -
(a) ~~60 MPa~~
(b) ~~50 MPa~~
(c) **80 MPa**
(d) None of these
- (ii) Hook's Law holds good for upto-
(a) Elastic Limit.
(b) Plastic Limit
(c) Proportional Limit.
(d) None of the above.
- (iii) Torsional rigidity is given by-
(a) GJ
(b) G/J
(c) $G\theta$
(d) T/J
- (iv) For Uniformly Distributed Load over a cantilever beam, the SF Diagram is-
(a) Straight line
(b) Parabolic
(c) Cubic
(d) None of the above
- (v) Euler's Buckling theory is applicable for -
(a) Short column
(b) Long column
(c) Medium long column
(d) All of them

2. Draw shear force and bending moment diagram for the cantilever beam shown in *Fig. 1*.

[14]

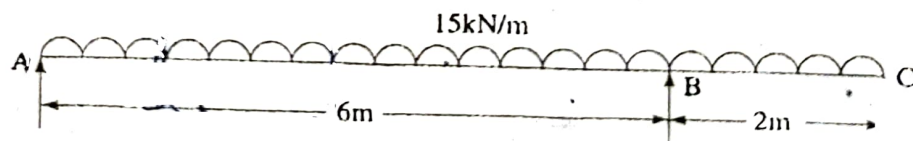


Fig.1

[Turn Over]

3. The state of stress in a material stressed to two-dimensional state of stress is as shown in *Fig. 2*. Determine principal stresses and maximum shear stress and the planes on which they act. [14]

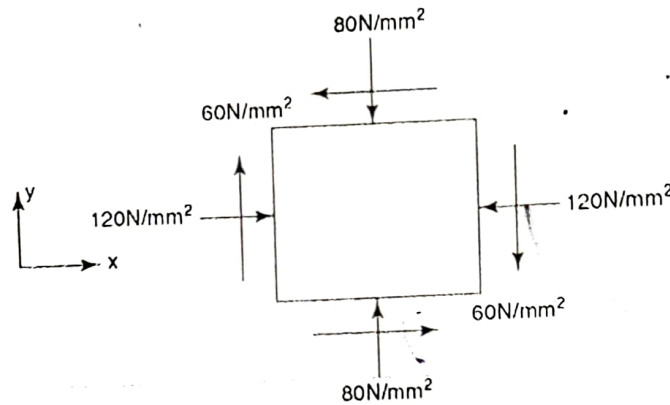


Fig. 2

4. Deduce the expression of Euler's Critical load for the column whose both ends are fixed. [14]
5. Three steel plates are welded to form a column section shown in *Fig. 3*. If the effective length of the column is 7.5 m, find the factor of safety with respect to buckling for a centric load of 12 kN. Take, $E = 2 \times 10^5 \text{ N/mm}^2$. [14]

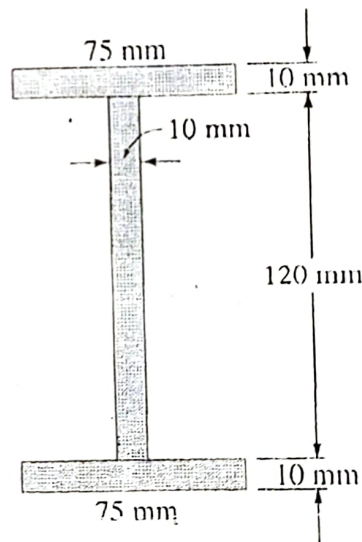


Fig. 3

6. Prove that a hollow shaft is stronger and stiffer than the solid shaft of the same material, length and weight. [14]
7. (a) What is slenderness ratio ?
 (b) What is Rankine's Constant in case of a column & strut ? Explain it.
 (c) What do you mean by Principal Plane and Principal Stress ?
 (d) What are the assumptions that have made to deduce the Euler's Critical load for column and strut.

[2+5+5+2]

*****THE END*****

[Turn Over]